Phol Castañeda Henao

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Junior Full-Stack Developer with a strong interest in scalable web application development. I have experience in academic and personal projects involving REST APIs, frontend and backend integration, and the design, administration, and optimization of relational databases, applying Clean Code principles and programming best practices. I stand out for my fast learning ability, analytical thinking, problem-solving skills, and teamwork. I am currently in the 8th semester of Systems Engineering at the University of Medellín, where I have built a solid academic foundation in areas such as software development, databases, agile methodologies, Scrum, software architectures, requirements engineering, user stories, and software testing.

SKILLS – TOOLS AND TECHNOLOGIES

Programming Languages: JavaScript · TypeScript · Java · Python
Frontend: HTML · CSS · React · Tailwind CSS · Next.js
Backend: Node.js · NestJS · Spring · Spring Boot · REST APIs
Databases: SQL · PostgreSQL · MySQL · SQL Server · SQLite
Other Tools: Git · Postman · Docker · CI/CD · VSCode · SonarQube

EXPERIENCE – PROJECTS

FluxLab
Web-based LIMS platform for managing clients, projects, and biotechnology samples.

- Developed a full-stack solution for the Biotechnology Laboratory at the University of Medellín, using a NestJS backend and a React frontend to centralize clients, projects, samples, and reports into a unified operational platform.
- Implemented sample templates with dynamic fields, data types, and field ordering to ensure consistency and traceability.
- Reduced the average sample registration time by 40% through bulk sample uploads with values in a single transaction and duplicate detection per project.
- Developed dashboards with metrics, recent activity, and workflow statuses, as well as Excel-exportable reports with summaries by template and status to facilitate decision-making and result delivery.
- Applied role-based access control and integrated JWT authentication through Supabase for security and compliance.
- Demo: https://fluxlab-front.vercel.app/

FocusHub
Web application for managing tasks, events, focus sessions, and productivity statistics.

- Designed and developed the complete productivity workflow using a NestJS backend and an Angular frontend, including task management with statuses, deadlines, duplication, and status-based filters.
- Built a productivity module with a calendar (month/week/day) and focus techniques such as Pomodoro or custom sessions, allowing users to create events, view tasks by date, and manage timers with state persistence.
- Ensured secure user access with JWT authentication and protected endpoints, using TypeORM with PostgreSQL support.
- Created statistics dashboards with charts for focus hours, completed tasks, and trends, along with a daily progress panel.
- Demo: https://focushub-eight.vercel.app/

BiTrimURL
Full-stack URL shortener built with React, Node.js, and PostgreSQL.

- Designed a REST API to create and retrieve shortened URLs with custom or automatically generated codes, including format and length validations.
- Implemented direct redirection through URL codes and click tracking to provide basic metrics without requiring user accounts.
- Persisted links and metadata in PostgreSQL using parameterized queries and differentiated error handling (400/404/500).
- Built a UI with React, TypeScript, and Tailwind that validates inputs, supports custom codes, and displays real-time results while keeping recent links available and consistent.

EDUCATION AND CERTIFICATIONS

UNIVERSITY OF MEDELLÍN **Medellín, Colombia**
Systems Engineering (Currently Enrolled – 8th Semester out of 10)

PLATZI

- Java Spring Boot Certification
- Backend Development with NestJS Certification
- Backend Development with ExpressJS Certification
- PostgreSQL Certification

LANGUAGES

- Spanish (Native)
- English B2 (CEFR): English Certification, University of Medellín (400 hours)

ADDITIONAL SKILLS

- Problem-solving and analytical thinking
- Teamwork and technical communication
- Programming best practices and Clean Code
- Continuous learning